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Class 10 Biology

Chapter 6

Short Questions Notes

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Chapter 6 – Animal Reproduction

Section II: Short Answer Questions

1. What is the primary purpose of reproduction in living organisms?

The primary purpose of reproduction is to ensure the continuation and survival of a species by producing new individuals of the same kind. It allows genetic material to be transferred from parents to offspring, maintaining the population size and introducing genetic variation that helps species adapt to environmental changes.

2. What are the two main types of reproduction?

Reproduction is mainly of two types: **Asexual reproduction** and **Sexual reproduction**.

In **asexual reproduction**, only one parent is involved, and the offspring are genetically identical to the parent, such as in Hydra or Amoeba.

In **sexual reproduction**, two parents (male and female) contribute gametes, resulting in genetically diverse offspring. This diversity is essential for evolution and adaptation.

3. At what stage of human development does the reproductive system become functional?

The reproductive system becomes functional during **puberty**, a stage that begins in the teenage years. During puberty, hormonal changes trigger the development of secondary sexual characteristics and activate the reproductive organs — in boys, testes start producing sperm, and in girls, ovaries begin releasing eggs and the menstrual cycle starts.

4. Which hormone is primarily responsible for the development of male secondary sexual characteristics?

The hormone **testosterone**, produced by the **Leydig cells** in the testes, is responsible for male secondary sexual characteristics such as facial hair growth, deepening of the voice, development of muscles, and increased bone mass. It also plays a vital role in sperm production and maintaining male reproductive health.

5. What role do Sertoli cells play in the male reproductive system?

Sertoli cells are specialized cells found inside the seminiferous tubules of the testes. They provide **nourishment, support, and protection** to developing sperm cells during spermatogenesis. They also help in the removal of damaged sperm and secrete certain substances that regulate the process of sperm formation and maturation.

6. Which hormone is essential for spermatogenesis and acts on Sertoli cells?

The hormone **Follicle-Stimulating Hormone (FSH)** is essential for spermatogenesis. It acts directly on **Sertoli cells** to stimulate and maintain sperm production. FSH works together with **testosterone**, secreted under the influence of **Luteinizing Hormone (LH)**, to ensure a continuous and healthy production of sperm.

7. Which hormone is primarily responsible for the development of female secondary sexual characteristics?

The hormone **estrogen**, secreted mainly by the ovaries, is responsible for the development of female secondary sexual characteristics such as breast enlargement, widening of hips, and regulation of the menstrual cycle. It also helps in the growth and maintenance of the uterine lining and overall female reproductive health.

8. What triggers ovulation in the female reproductive cycle?

Ovulation is triggered by a sudden rise or surge in **Luteinizing Hormone (LH)** in the middle of the menstrual cycle. This hormone causes the mature egg to be released from the ovary into the fallopian tube, where it can be fertilized by a sperm cell. This LH surge is a key event that marks the fertile phase of the cycle.

9. Where is progesterone produced in the female reproductive system after ovulation?

After ovulation, **progesterone** is produced by a temporary structure in the ovary called the **corpus luteum**. This hormone prepares the uterus for possible pregnancy by thickening the uterine lining. If fertilization does not occur, progesterone levels drop, leading to menstruation.

10. What is the term for the process of sperm cell production?

The process of sperm cell production is called **spermatogenesis**. It takes place in the seminiferous tubules of the testes and involves the formation of mature, motile sperm from germ cells through cell division and differentiation. It is a continuous process that begins at puberty and continues throughout a male's life.

11. In which part of the male reproductive system does spermatogenesis occur?

Spermatogenesis occurs inside the **seminiferous tubules** located within the **testes**. These tubules contain germ cells that divide and mature into sperm, supported by Sertoli cells. The mature sperm are then transported to the epididymis for storage and further development.

12. Where does fertilization typically occur in the female reproductive system?

Fertilization usually occurs in the **fallopian tube** (also known as the oviduct). This is where the sperm meets and fuses with the egg to form a zygote. The zygote then travels to the uterus for implantation and further development into an embryo.

13. What is the genetic outcome of offspring produced through asexual reproduction?

Offspring produced through asexual reproduction are **genetically identical** to the parent, meaning they are clones. This is because there is no mixing of genetic material from two parents. Although this ensures stability, it limits genetic variation and adaptability to changing environments.

14. What role does the scrotum play in the reproductive system of a male rabbit?

The **scrotum** is a pouch-like structure that holds and protects the testes outside the body cavity. Its main function is to regulate the temperature of the testes, keeping it slightly lower than body temperature. This cooler environment is essential for the proper development and survival of sperm cells.

15. Which part of the female reproductive system in rabbits is the typical site for fertilization?

In rabbits, **fertilization takes place in the oviduct (fallopian tube)**. When mating occurs, sperm travel up the female's reproductive tract and meet the ovum in the oviduct, where fusion happens to form a zygote. The zygote then moves to the uterus for implantation.

Section III: Extensive Answer Questions

1. Explain the role of hormones in male sexual development, detailing the specific functions of testosterone and follicle-stimulating hormone (FSH). How do these hormones interact to regulate spermatogenesis?

Hormones play a crucial role in controlling male sexual development and reproduction. The brain's **pituitary gland** releases two important hormones: **FSH (Follicle-Stimulating Hormone)** and **LH (Luteinizing Hormone)**.

- **LH** acts on the **Leydig cells** of the testes to produce **testosterone**, the main male sex hormone. Testosterone is responsible for the development of male secondary sexual characteristics such as a deeper voice, beard growth, and increased muscle mass. It also stimulates sperm production.
- **FSH** acts on **Sertoli cells** in the seminiferous tubules, helping in the nourishment, growth, and maturation of developing sperm.
Together, FSH and testosterone regulate **spermatogenesis**, ensuring that healthy and mature sperm are continuously produced. This hormonal coordination maintains fertility and male reproductive health.

2. Compare and contrast the processes of spermatogenesis and oogenesis.

Spermatogenesis and **oogenesis** are both processes of gamete formation through meiosis, but they differ in timing, outcome, and location.

- **Spermatogenesis** occurs in the **seminiferous tubules of the testes** and starts at puberty. It continues throughout a male's life, producing millions of sperm daily. Each germ cell divides to form four functional sperm cells.
- **Oogenesis**, on the other hand, occurs in the **ovaries** of females and begins before birth. It pauses until puberty, after which one egg matures during each menstrual cycle. Unlike spermatogenesis, oogenesis produces only one functional ovum and three small polar bodies that degenerate.
While both involve meiosis to reduce chromosome number, spermatogenesis ensures quantity, and oogenesis ensures quality, producing a nutrient-rich egg for fertilization.

3. Discuss the process of fertilization in humans. How does fertilization restore the diploid chromosome number?

Fertilization in humans is the process by which a sperm cell from the male fuses with an ovum from the female to form a **zygote**. It takes place in the **fallopian tube**. When a sperm successfully penetrates the egg's membrane, their nuclei combine.

The sperm and egg each contain **23 chromosomes** (haploid). When they fuse, the resulting

zygote has **46 chromosomes** (diploid), restoring the normal chromosome number.

This union not only ensures species continuity but also introduces **genetic variation**, since genes from both parents combine to form a unique individual. Fertilization marks the beginning of embryonic development.

4. Outline the key steps involved in sexual reproduction using the rabbit as an example.

In rabbits, sexual reproduction involves several coordinated steps:

1. **Gamete Formation:** The male produces sperm in the testes, while the female produces eggs in the ovaries through gametogenesis.
2. **Mating:** During copulation, sperm are transferred from the male to the female reproductive tract.
3. **Fertilization:** Occurs in the **oviduct**, where the sperm and egg fuse to form a zygote.
4. **Development:** The zygote divides and develops into an embryo, which implants into the uterus wall.
5. **Birth:** After a short gestation period, the female gives birth to live young.
This process ensures that each generation of rabbits inherits a mix of genetic traits, increasing adaptability and survival chances.

5. Describe the mechanisms of asexual reproduction in animals.

Asexual reproduction is a mode of reproduction that requires only one parent, producing genetically identical offspring.

Different mechanisms include:

- **Binary Fission:** The parent cell divides into two equal parts, as in Amoeba or Paramecium.
- **Budding:** A small bud grows from the parent's body and detaches to form a new organism, as seen in Hydra.
- **Fragmentation:** The body breaks into fragments, each capable of forming a new individual, as in Planaria.
- **Regeneration:** Lost body parts can regrow into complete organisms, such as in starfish.
- **Parthenogenesis:** Offspring develop from unfertilized eggs, as seen in some insects and reptiles.
These methods are faster and energy-efficient but produce little genetic diversity.

6. Explain the process of sex determination in humans.

In humans, sex is determined by the combination of **sex chromosomes**. Females have **XX** chromosomes, and males have **XY**. The mother's egg always contributes an X chromosome, while the father's sperm carries either X or Y.

If the sperm carrying an **X** fertilizes the egg, the result is a **female (XX)**. If it carries a **Y**, the child will be **male (XY)**.

The **SRY gene**, present on the Y chromosome, triggers the formation of testes and secretion of testosterone, which leads to male development. Therefore, it is the sperm that ultimately determines the sex of the child.

7. Discuss the differences between viviparous, oviparous, and ovoviviparous animals.

Animals reproduce differently based on how their offspring develop:

- **Viviparous animals** (like humans, rabbits, and cats) give birth to live young. The embryo develops inside the mother's body and is nourished through the placenta.
- **Oviparous animals** (like birds, frogs, and fish) lay eggs outside the body. The embryo develops within the egg and relies on stored nutrients.
- **Ovoviviparous animals** (like some sharks and reptiles) produce eggs that hatch inside the mother, and the young are born live.
Each method suits the animal's habitat and ensures maximum survival according to environmental conditions.

8. Explain the significance of gametogenesis in sexual reproduction.

Gametogenesis is the formation of male (sperm) and female (egg) gametes through **meiosis**. It is crucial because it reduces the chromosome number from diploid (46) to haploid (23), ensuring that when fertilization occurs, the zygote regains the correct diploid number.

Moreover, meiosis introduces **genetic variation** through crossing over and random assortment of chromosomes. This variation makes each individual genetically unique, strengthening a species' ability to adapt, evolve, and survive changes in the environment.